Renewable-Energy PoC — Technical Documentation

Medallion lakehouse on Databricks | updated 2026-07-28

1. Architecture

Databricks Asset Bundle with DLT (Delta Live Tables) pipelines on serverless, Auto Loader (cloudFiles) ingestion from Unity Catalog Volumes, and Unity Catalog governance. Flow: Sources → Ingestion → Bronze → Silver → Gold → Consumption, orchestrated by Databricks Jobs and monitored with Lakehouse Data-Quality Monitors.

!Architecture

!End-to-end data flow

!End-to-end workflow

2. Domains & data sources

- Weather — solar irradiance + wind resource. Sources by role: NASA POWER = historical hourly (measured-equivalent) + a 12-city world daily benchmark, DMC = real-time measured stations, MinEnergía/DGF = modeled resource (typical year / 1980–2017).
- Resource (renewable_energy_chile generation): per-plant generation coordinado/real/reducciones. Source: CEN / Coordinador. Silver = Data Vault (hub/link/sat).
- Conglomerate (renewable_conglomerate_energy): country stats (Entity/Year). Source: OWID.

3. Weather medallion — dedicated schemas (dev catalog)

All weather is officialized under three schemas, each holding per-source tables plus one consolidated all_sources table (a source column tags every row):

Layer	Schema	Tables
Bronze	dev.weather_bronze	nasa_power, dmc, dgf, all_sources (882,317)
Silver	dev.weather_silver	nasa_dim_plant, nasa_fact_hourly, dmc_dim_station, dmc_fact, dgf, all_sources (881,602)
Gold	dev.weather_gold	nasa_resource_kpi, nasa_daily, nasa_monthly, dmc_latest, dmc_station_kpi, dgf_resource_kpi, all_sources_kpi, + dashboard-serving tables chile_daily_2025_2026, world_daily_2023_2024, world_daily_2025_2026

Schemas and key tables carry Unity Catalog COMMENTs (self-documenting); naming is symmetric by region/period. The consolidated all_sources shape is source, location, location_type, resource, lat, lon, momento, ghi, temperatura, humedad, presion, agua_caida, viento, direccion_viento.

4. Data acquisition (fetchers → UC Volume → Auto Loader)

Fetcher (scripts/)	Source	Endpoint / access	Notes
nasa_power_fetcher.py	NASA POWER	power.larc.nasa.gov/api/temporal/hourly/point (no auth)	historical hourly by lat/lon; yearly chunks; DMC-equivalent fields

dmc_ema_fetcher.py	DMC / meteochile	getDatosRecientesEma/<code>?usuario&token	real-time measured, last 12 h, per-minute
dgf_explorador_fetcher.py	DGF Explorador	python-router (no auth)	modeled typical-year
minenergia_api_fetcher.py	MinEnergía official API	POST /api/proxy (session cookie)	hourly 1980–2017
cen_coordinador_fetcher.py	CEN / Coordinador	sipubv1.api.coordinador.cl/api/v1 (public key)	generation per plant; public by law
owid_conglomerate_fetcher.py	OWID	grapher CSVs (no auth)	5 country datasets

5. Weather — NASA POWER load & world benchmark

- Historical (Chile): hourly ALLSKY_SFC_SW_DWN (GHI), T2M, RH2M, PS, PRECTOTCORR, WS10M/WD10M/WS50M per plant coordinate, in yearly chunks. Full load: 876,720 rows (10 plants × 87,672 hours, 2015–2024) → UC Volume → bronze → silver star schema (nasa_dim_plant + nasa_fact_hourly, @expect_all_or_drop) → gold marts (nasa_resource_kpi with rank + A/B/C tier, nasa_daily 36,530, nasa_monthly 1,200). Physically validated: midday GHI highest for the Atacama plants (~871 W/m²).
- World benchmark: NASA POWER daily for 12 cities across all continents (Santiago, São Paulo, New York, Los Angeles, London, Berlin, Cairo, Nairobi, Dubai, Mumbai, Tokyo, Sydney), landed in world_daily_2023_2024 (8,772) and world_daily_2025_2026 (6,852). Northern-vs-Southern hemisphere inversion verified (London Jan 4 °C / Jul 17 °C vs Sydney Jan 23 °C / Jul 14 °C).

6. Consumption — AI/BI dashboards (Lakeview)

- Weather — Chile & Rest of World (01f18a95…): one dashboard, two tabs. Seasonality & trends, max/min/means of temperature, GHI, pressure, wind; per-location and per-city breakdowns; hemisphere seasonality bar.
- Weather — Chile & Rest of World, 2025–2026 YTD (01f18a9c…): the same views on the latest data.
- Weather NASA (01f18771…): time series + resource-KPI table.
- Lakeview note: charts render with disaggregated:false + aggregation in the field expression; tables need disaggregated:true + a pre-aggregated dataset.

7. Data quality, contracts & observability (Databricks-native)

- Contracts: documented per medallion step in ../data-contracts.md.
- In-pipeline DQ: DLT @dlt.expect_* on the silver layers (Chile-bounds coords, known resource, has-metric, valid keys/dates) + UC constraints.
- Lineage & glossary: Unity Catalog automatic lineage + column/table comments; ../glossary.md.
- Monitoring: three Data-Quality Monitors on the gold tables (profiling + drift, each with an auto-generated dashboard) + a SQL alert that fires on failed expectations.

8. Deploy & run

```
databricks bundle deploy -t dev --var=dev_catalog=workspace
databricks bundle run weather_streaming_job -t dev --var=dev_catalog=workspace # Weather
databricks bundle run conglomerate_owid_job -t dev --var=dev_catalog=workspace # Conglomerate (OWID)
databricks bundle run cen_generation_job -t dev --var=dev_catalog=workspace # Resource (CEN)
```

All work is done in dev and on branches. Prod deploys are managed by the project lead (prod target scoped to their workspace). Note: deploying from a branch that lacks a resource deletes it in dev — deploy from a branch that contains everything, or merge first.

9. Verify the medallion, jobs & dashboards

- Weather — Chile & Rest of World (2 tabs): https://dbc-54b27bae-2e91.cloud.databricks.com/sql/dashboardsv3/01f18a95d5851bb1a33415a71141a908?o=7474645896934260
- Weather — Chile & Rest of World, 2025–2026 YTD: https://dbc-54b27bae-2e91.cloud.databricks.com/sql/dashboardsv3/01f18a9c8d2f19a48485521da3f5a4f6?o=7474645896934260
- Weather NASA — dashboard (series + KPIs): https://dbc-54b27bae-2e91.cloud.databricks.com/sql/dashboardsv3/01f18771593d19568e23e322277007c3?o=7474645896934260
- Weather gold schema: https://dbc-54b27bae-2e91.cloud.databricks.com/explore/data/dev/weather_gold?o=7474645896934260
- Resource pipeline: https://dbc-54b27bae-2e91.cloud.databricks.com/pipelines/9bcfc32a-19f0-4327-b5bc-e8ee1cc4a3a4?o=7474645896934260
- Conglomerate pipeline: https://dbc-54b27bae-2e91.cloud.databricks.com/pipelines/b5d63282-a439-42ef-8e3b-cd707a9d5f58?o=7474645896934260
- Weather pipeline: https://dbc-54b27bae-2e91.cloud.databricks.com/pipelines/c108b287-98cc-43d6-86d6-5b63a9e6b4ae?o=7474645896934260
- Quality monitors: Resource · Weather · Conglomerate